# Basic Mechanical Engineering
# ~ ME 101 ~

Semester: Spring 2016

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National University of Computer and Emerging Sciences
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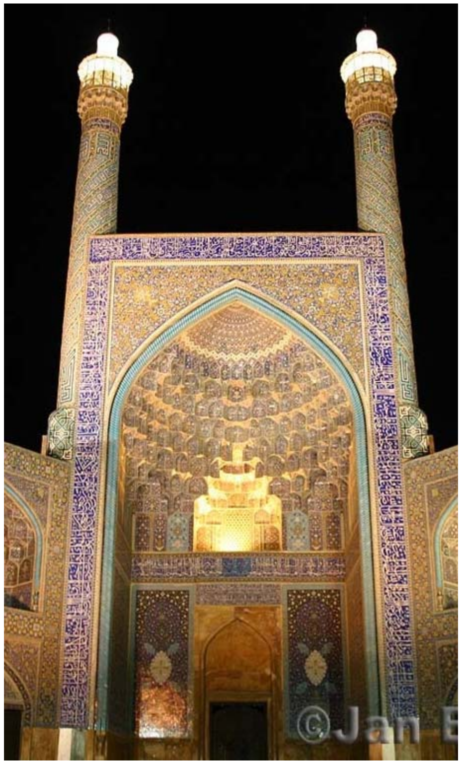
# Introduction

- Instructor
- Basic Mechanical Engineering
- Introduction

# Instructor

## PERSONAL

**~ Born ~**
**Lahore, Pakistan**



**~ Raised ~**
**Isfahan, Iran**

**~ Seasoned ~**
**Paris, France**

# Instructor

## EDUCATION

**University of Eng. & Technology**
**Lahore, Pakistan**


**B.Sc. (Hons)**
**Mechanical Engineering**
**[1998 - 2002]**

**National University, FAST**
**Lahore, Pakistan**

**Masters**
**Computer Science**
**[2002 – 2005]**

**Arts et Métiers ParisTech**
**Paris, France**

**Masters + PhD**
**Mechanical Engineering**
**[2005 - 2010]**

# Instructor

## EXPERIENCE

**Arts et Métiers ParisTech**
**Paris, France**

**Research and Teaching Attaché**
**PIMM Laboratory**
**[2009 - 2010]**

**University of Engg. & Technology**
**Lahore, Pakistan**

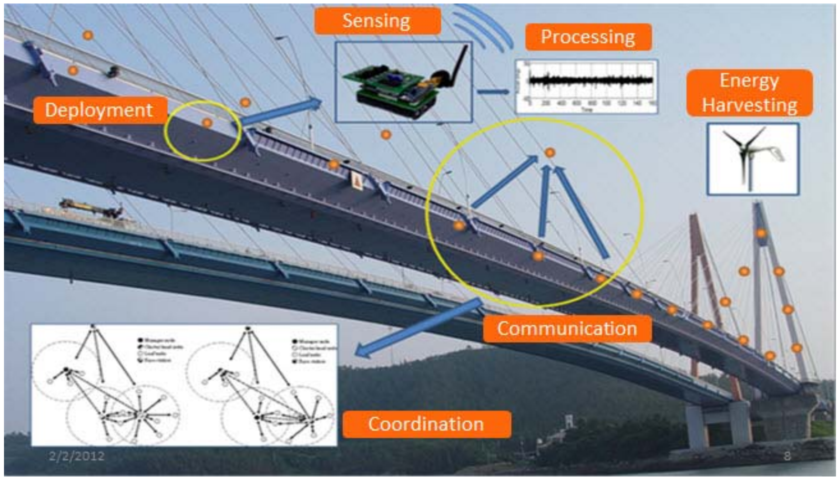

**Assistant Professor**
**Mechatronics Department**
**[2010- 2012]**

**King Abdulaziz University,**
**Jeddah, Saudi Arabia**


**Assistant Professor**
**Mechanical Engineering Dept.**
**[2012 - 2014]**

# Instructor

## Research Interests



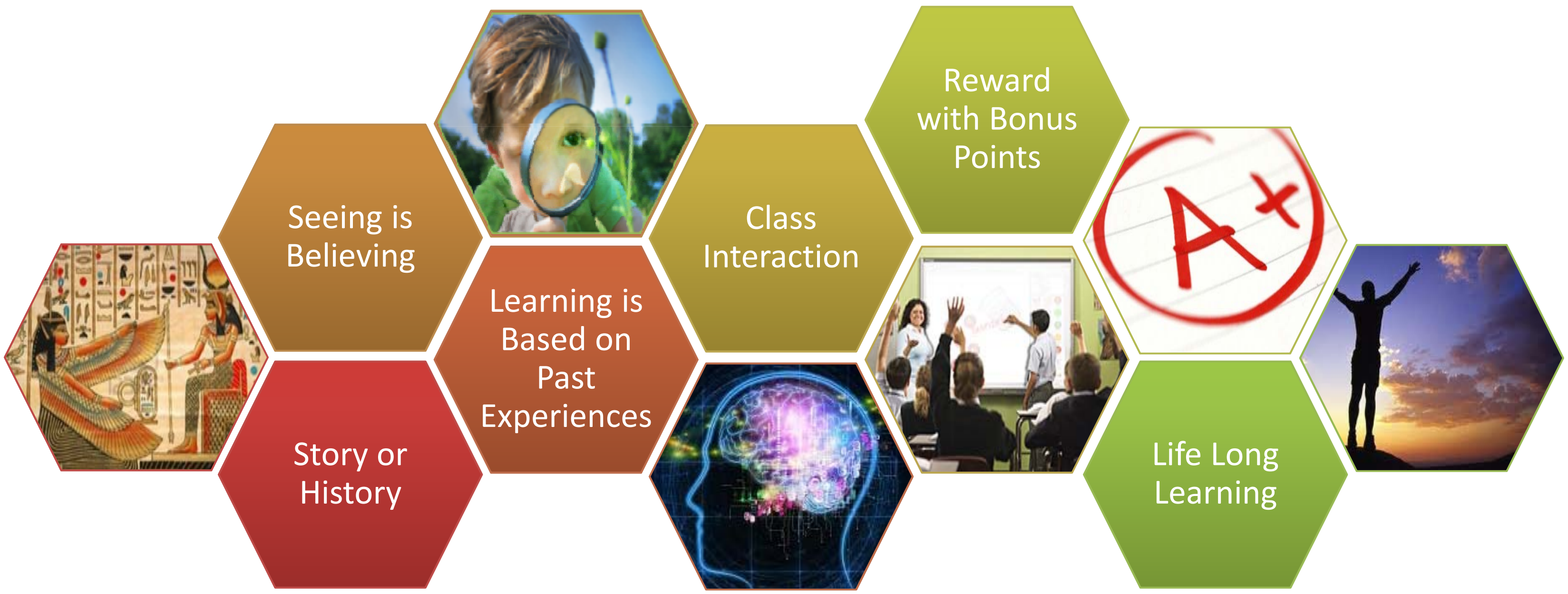

**Structural Health Monitoring**

**System Identification**

**Robotics**

# Instructor

## TEACHING METHODOLOGY



# Instructor

## SLIDE THEME

**Questions, Disadvantages &Warnings**

**Answers & Advantages**

**Definitions & Facts**

[1] - References

# Basic Mechanical Engineering

## IDEA

**What does "Engineering" Mean?**

**Application of scientific, economic, social, and practical knowledge in order to invent, design, build, maintain, research, and improve structures, machines, devices, systems, materials and processes.**

**What is Mechanical Engineering?**

**Discipline that applies the principles of engineering, physics and materials science for the design, analysis, manufacturing, and maintenance of mechanical systems**

**What is meant by Mechanical Systems?**

**A mechanical system manages power to accomplish a task that involves forces and movement.**

# Basic Mechanical Engineering

## ENGINEER

Cleverness

**What does an Engineer do?**

**Applies scientific knowledge and ingenuity to develop solutions for technical, societal and commercial problems**


**What are you doing here?**

**Learning scientific knowledge and its applications**

**YOU ALL ARE PROBLEM SOLVERS**

# Basic Mechanical Engineering

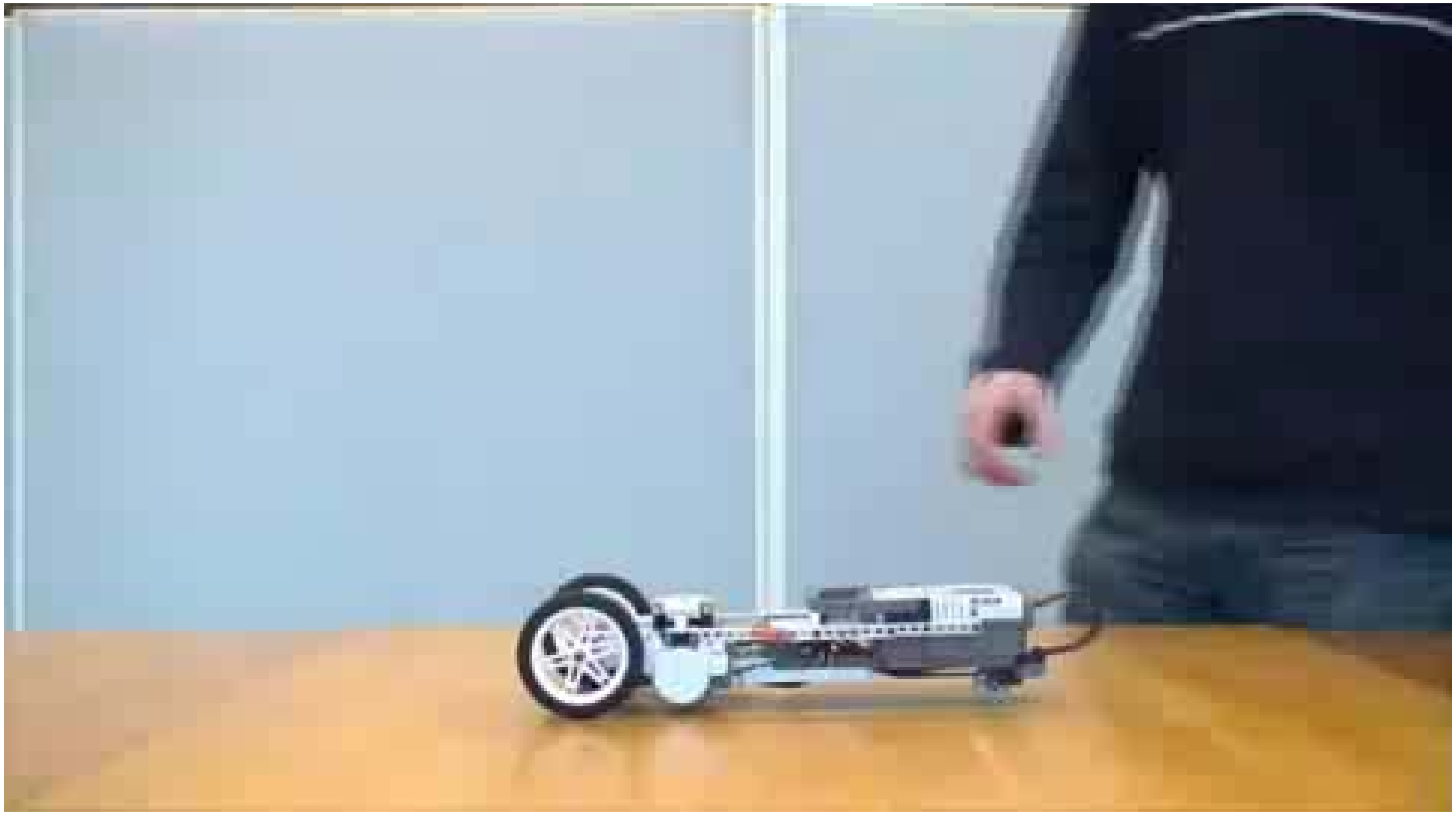
## Applications


**The Astounding Athletic Power of Quad copters**

[1] - Raffaello D'Andrea: The astounding athletic power of quadcopters, TEDGlobal (JUN 2013), URL :http://on.ted.com/Quadcopter

# Basic Mechanical Engineering

## APPLICATIONS

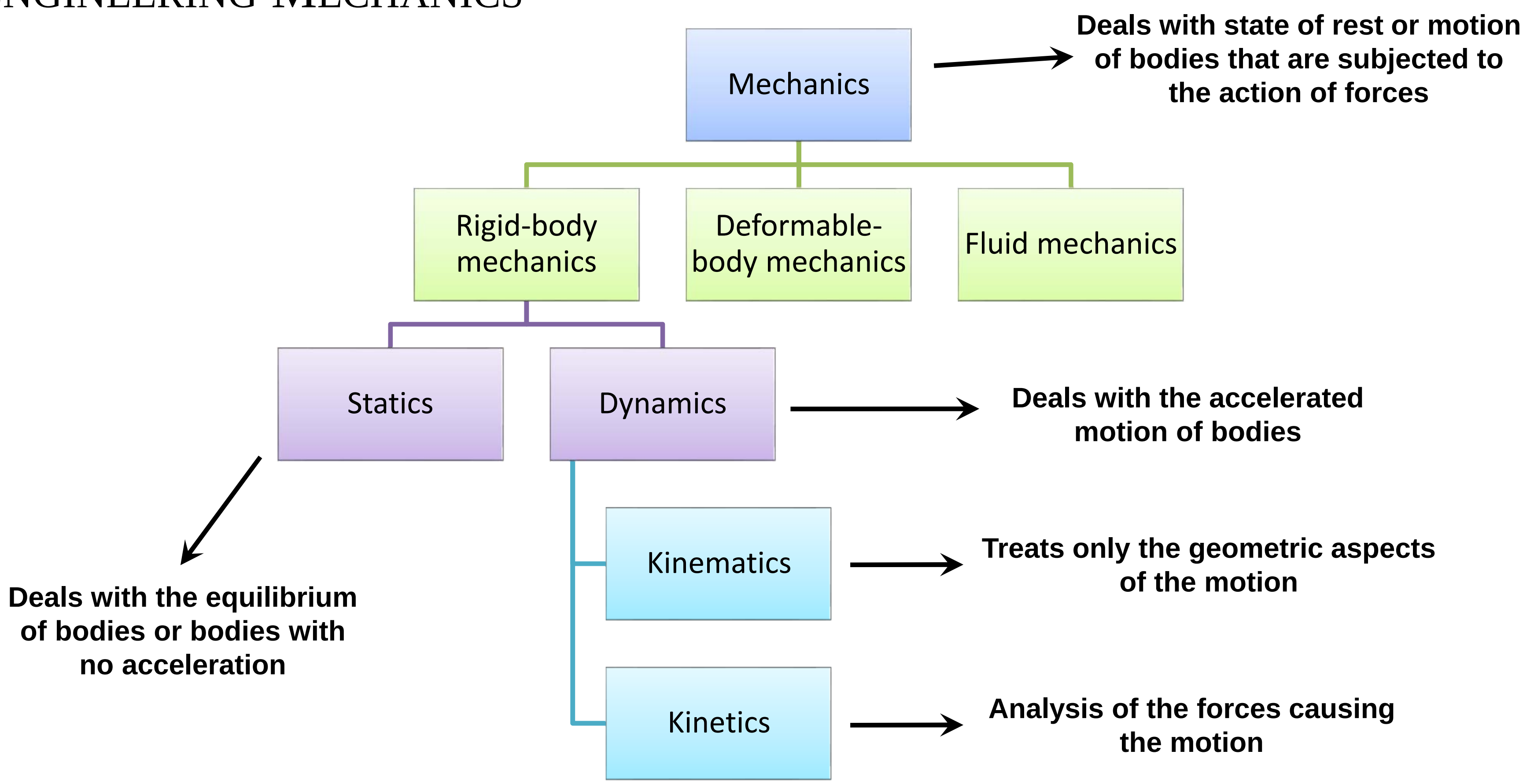


**Engineering Mechanics is the study of forces and movements** [1]

[1] – Youtube – January 1, 2015 (www.youtube.com/watch?v=RFFidyojDmk)

# Basic Mechanical Engineering

## Engineering Mechanics




**Essential for the design and analysis of many types of structural members, mechanical components, or electrical devices encountered in engineering**

# Basic Mechanical Engineering

## Objective

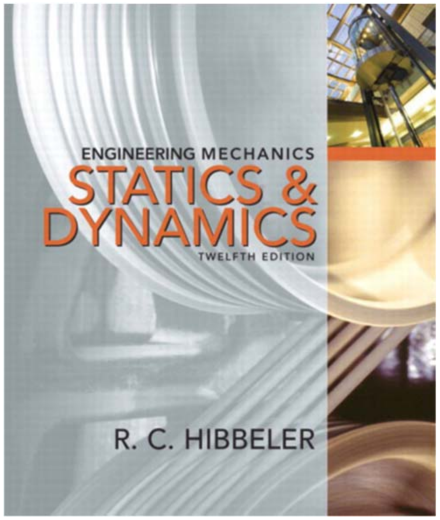
**To build the fundamental concepts of engineering mechanics and to clearly present their application in the analysis of planner motion of mechanical components and structures.**

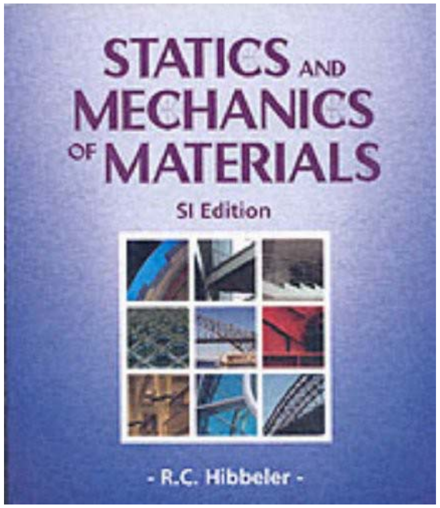
## Course Book

- ✓ *Engineering Mechanics: Statics and Dynamic*, by R.C Hibbeler, Pearson Prentice Hall, 12th Edition, 2010.

## Reference Books

- ✓ *Statics and Mechanics of Materials*, by R.C Hibbeler, Pearson Prentice Hall, SI Edition, 2004.
- ✓ *Engineering Mechanics: Statics and Dynamics*, by J. L. Meriam, and L. G. Kraige, John & Wiley Sons , 7th Edition,, 2012



# Basic Mechanical Engineering

## Course Outline

| Week Number | Book Topics Numbers | Topic Names |
|---|---|---|
| **1** | 1.1 to 1.3 | Introduction to Vector Mechanics |
| **2** | 2.1 to 2.7 | Force Systems |
| **3** | 3.1 to 3.2 | Free Body Diagram + **Quiz I** |
| **4** | 3.5 to 3.10 | Equilibrium of a Rigid Body |
| **5** | 5.1 to 5.2 | Friction |
| **6** | **Midterm Examination I** | |

***Appropriate changes would be made if required.***

# Basic Mechanical Engineering

## Course Outline

| Week Number | Book Topics Numbers | Topic Names |
|---|---|---|
| 7 | 16.1 to 16.4 | Planar Kinematics: Absolute Motion Analysis |
| 8 | 16.5, 16.7 | Planar Kinematics : Relative Motion Analysis |
| 9 | 17 | Planar Kinetics: Force and Acceleration |
| 10 | 17 | Planar Kinetics: Force and Acceleration + **Quiz II** |
| 11 | 18 | Planar Kinetics: Work and Energy |
| 12 | | **Midterm Examination II** |

***Appropriate changes would be made if required.***

# Basic Mechanical Engineering

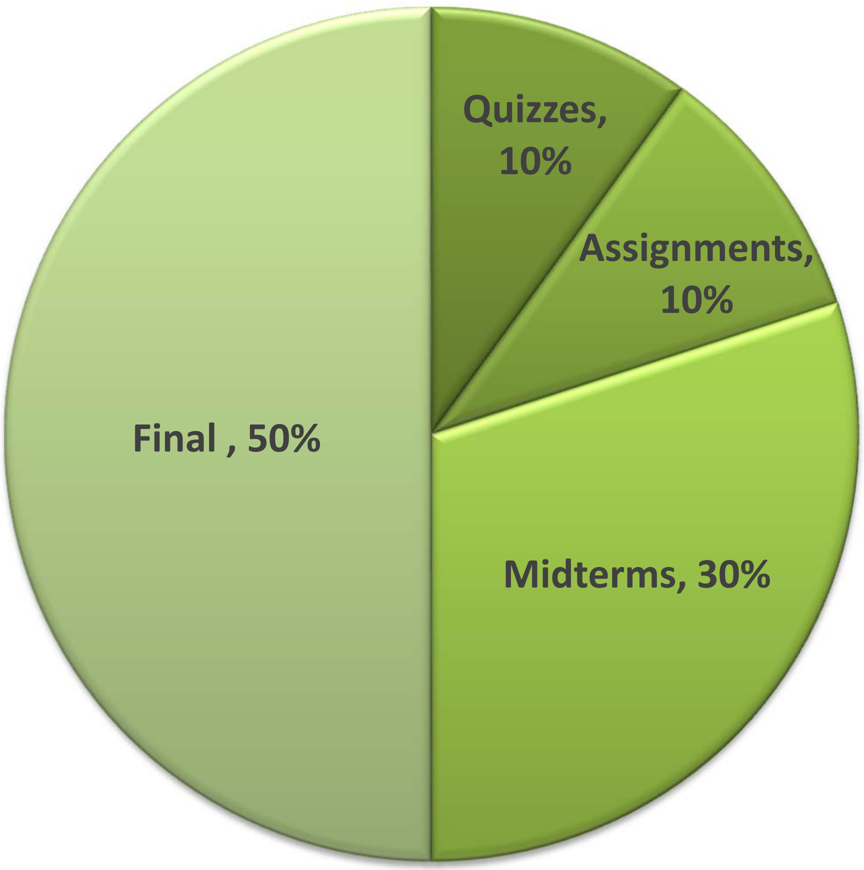
## Course Outline

| Week Number | Book Topics Numbers | Topic Names |
|---|---|---|
| **13** | 16.1 to 16.4 | Mechanics of Materials: Stress and Strain |
| **14** | 16.5, 16.7 | Mechanics of Materials: Torsion & Bending |
| **15** | 17 | Mechanical Properties of Material + **Quiz III** |
| **16** | 18 | Structure Analysis: Frames and Machines |
| | **Final Examination** | |

***Appropriate changes would be made if required.***

# Basic Mechanical Engineering

## Grading Policy




**Bonus, 5% Absolute (Max)**

# Basic Mechanical Engineering

## A+ OR F

**How to succeed in this course?**

- ✓ **Arrive on time**
- ✓ **Participate in all classes**
- ✓ **Pay attention, take notes and ask questions if needed**
- ✓ **Identify and understand key issues**
- ✓ **Review material (notes and textbook) before and after class**
- ✓ **Avoid Plagiarism**
- ✓ **Practice, Practice and Practice**

# Take-Home Message

## Thought

**You are a Problem Solver**

# Fundamental Concepts

- Basic Quantities
- Coordinate Systems

# Basic Quantities

## MEASURING OBJECTS

**How do we measure the size of a physical object?**

**Lengths & Angles**

**What are the units?**

**Meter - Radians**

**How long is 1 Meter?**

**One ten-millionth of the distance from the Earth's equator to the North Pole**

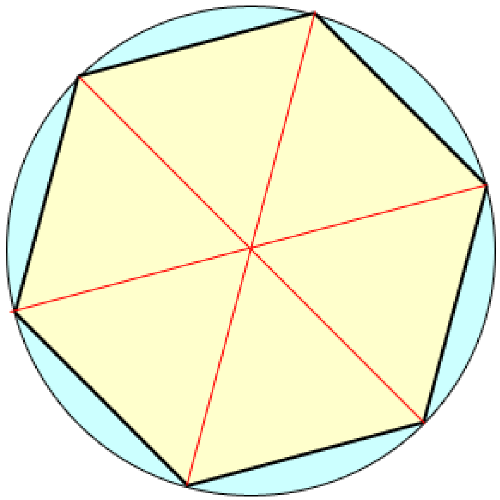
**Length of the path travelled by light in vacuum during a time interval of 1/299,792,458 of a second**

# Basic Quantities

## MEASURING OBJECTS

**Why a circle has 360 degrees?**
**Why not 248 or 623 degrees?**

**Babylonians subdivided the circle using the angle of an equilateral triangle as the basic unit and further subdivided the latter into 60 parts**



**Which one is greater degree or radian?**

**1 Radian = 57.3 Degrees**
**Radian > Degrees**

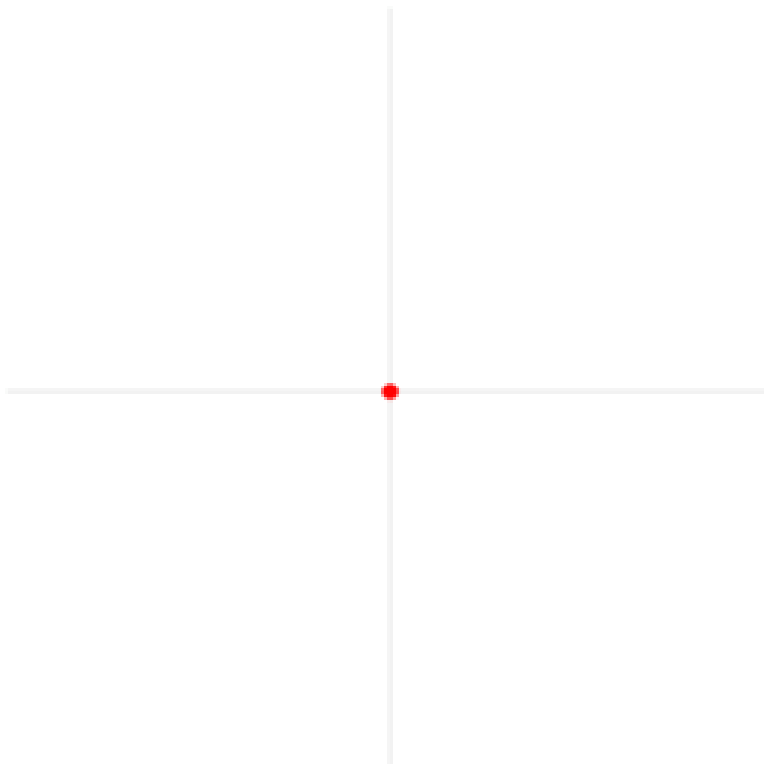
# Basic Quantities

## MEASURING OBJECTS

**How much is one Radian?**

**Numerically equal to the length of a corresponding arc of a unit circle**

**Length of arc ($l$) = Radius ($r$) x Angle ($\theta$)**



# Basic Quantities

## Radius of Earth

**What is the mean radius of earth?**

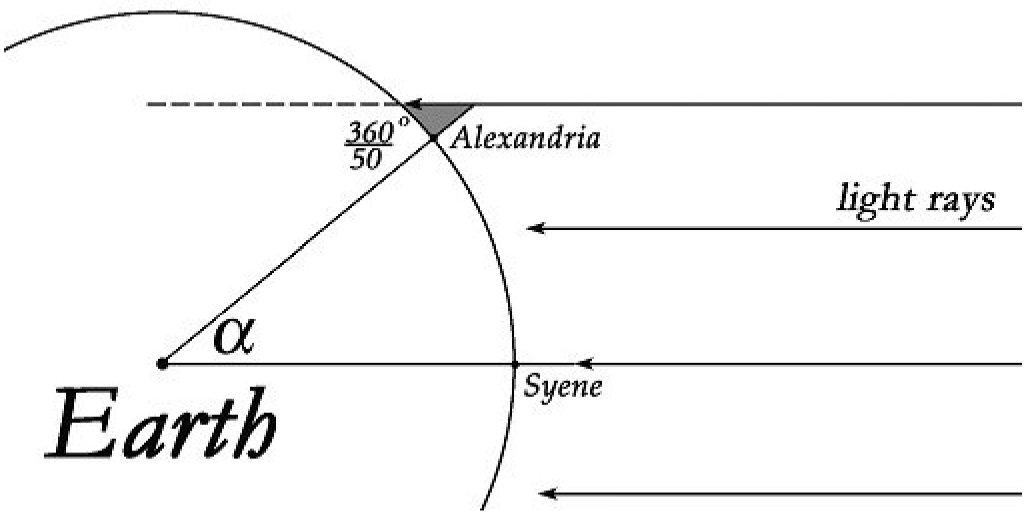
**6371 Km**

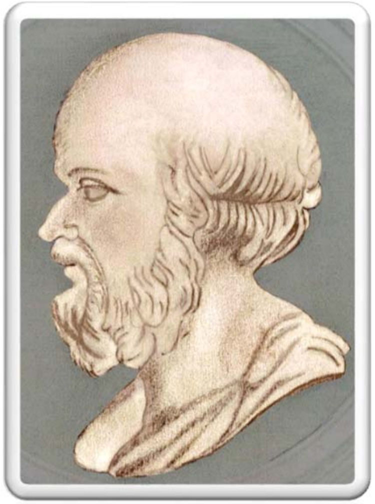
**Who calculated it?**

**Eratosthenes**

**How?**

$$l = r\,\theta$$



**Eratosthenes (276BC - 194BC) Greek mathematician, geographer, poet, astronomer**

# Basic Quantities

## Measurement Quantity

**How do we measure that amount of matter in an object?**

**Mass**

**Units?**

**Kilogram (Kg)**

**How is it defined?**
**How much is 1Kg?**

**1 gram is the mass of 1 cubic centimeter of water**

**How is it measured?**

**Inertial Balance**

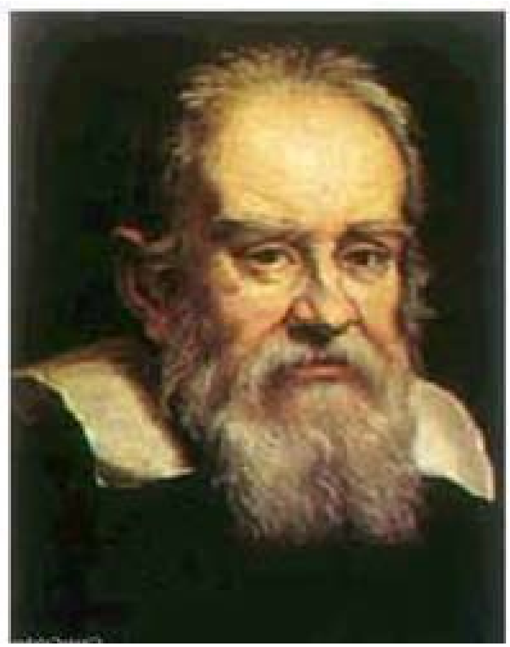
**What is the mass of earth?**

$$\mathbf{5.972 \times 10^{24}}\ \boldsymbol{Kg}$$

# Basic Quantities

## Mass of Earth

**How was mass of the Earth measured?**



**Galileo Galilei (1564-1642)**

$$g = 9.81\ m/s$$

**Isaac Newton (1642-1726)**

$$F = \frac{Gm_1m_2}{r^2}$$

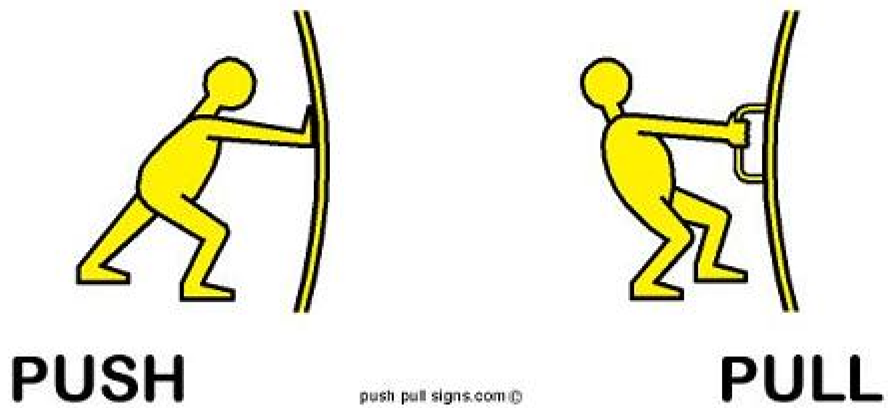
**Henry Cavendish (1731-1810)**

$$G = 66.73 \times 10^{-12} \frac{m^3}{Kg.s^2}$$

# Basic Quantities

## MEASURING PUSH/PULL

**Can we measure "Push" or "Pull"?**



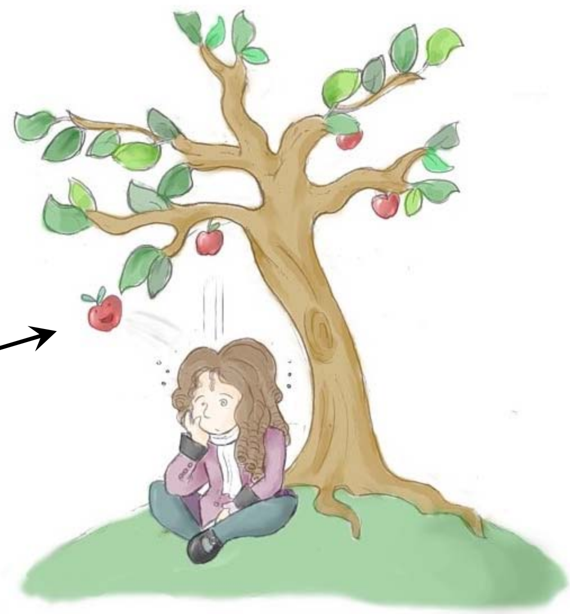

**Yes! Force in Newton**

# Basic Quantities

## MEASURING PUSH/PULL

**How much is the push/pull of one Newton force?**

**Apple of** $\approx 102\ g$



**What is the pulling effect of Earth on an object called?**

**Weight**

# Basic Quantities

## MEASURING PUSH/PULL

**Is weight of an object constant?**

**No!**

**Why?**

$$F = \frac{Gm_1m_2}{r^2}$$

$G$ **= Universal Constant of Gravitation =** $66.73 \times 10^{-12}$

# Coordinate Systems

## INTRODUCTION

**What do we need to specify the position of an object in 3D space ?**

**A reference**
**A convention for representation**

**What do we get by combining the above mentioned entities?**

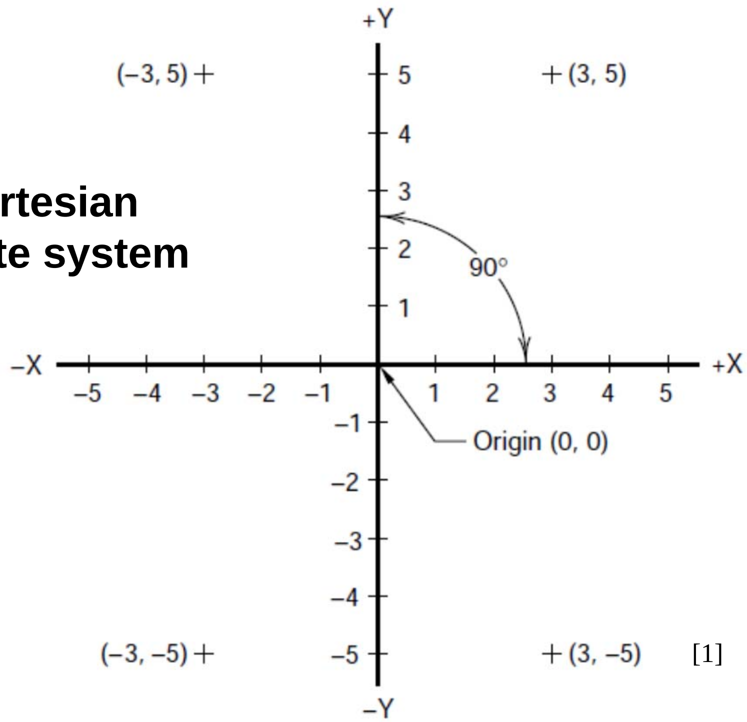
**A coordinate system**

# Coordinate Systems

## HISTORY

**for every point in space, a set of real numbers can be assigned, and for each set of real numbers, there is a unique point in space**

**2-D Cartesian coordinate system**




**Renè Descartes (1596–1650)**

**French Mathematician**

"*Cartesius*"

[1] - Gary Bertoline, Eric Wiebe et al, "Technical Graphics Communication", McGraw-Hill Science/Engineering/Math, 4 edition, January 31, 2008, p 307.

# Coordinate Systems

## HISTORY

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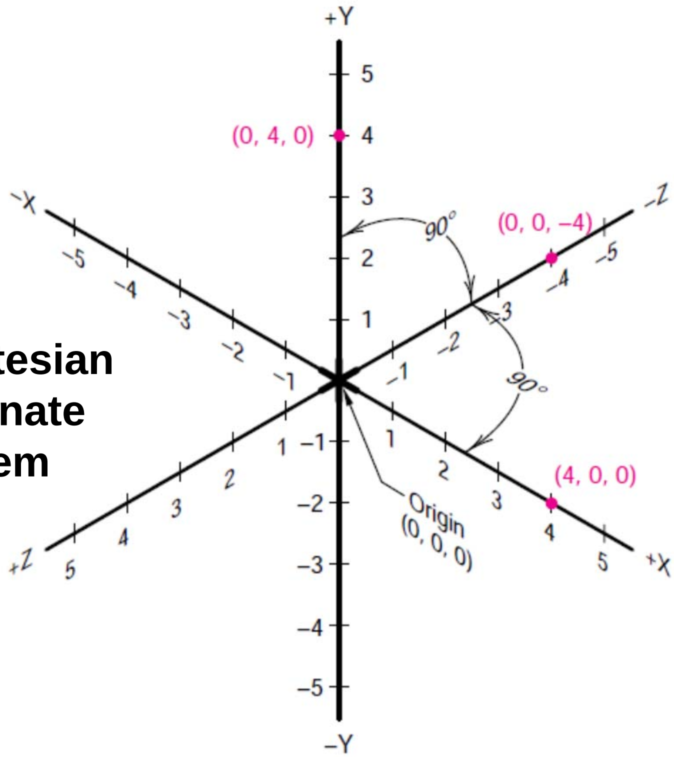


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**Renè Descartes (1596–1650)**

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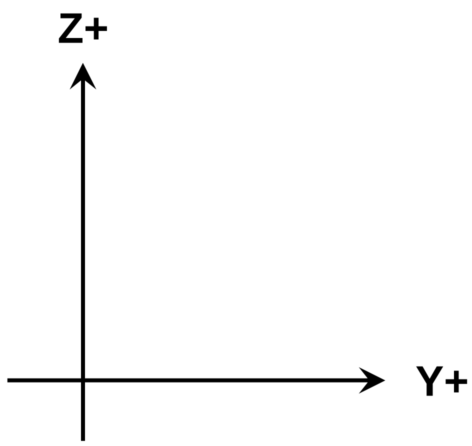
# Coordinate Systems

## CARTESIAN COORDINATE SYSTEM

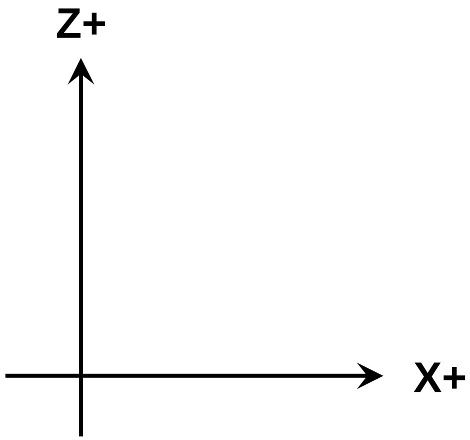



**In which direction is X+ axis?**
**In the plane or out of the plane?**

**Out of the plane**

# Coordinate Systems

## Cartesian Coordinate System




**In which direction is Y+ axis?**
**In the plane or out of the plane?**

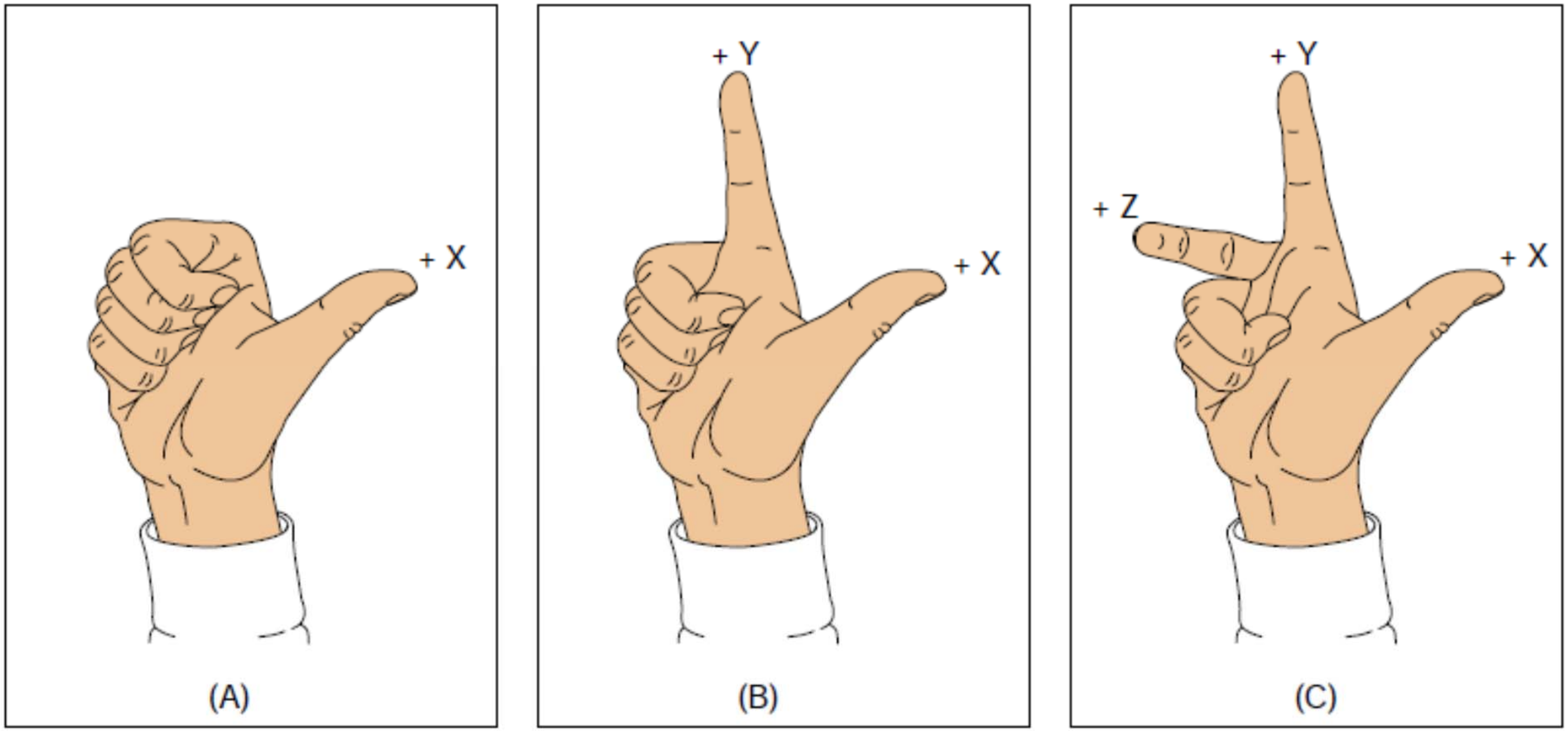
**In the plane**

**How do you know for sure?**
**What is the rule that you are applying?**

**Right Hand Rule**

# Coordinate Systems

## RIGHT HAND RULE



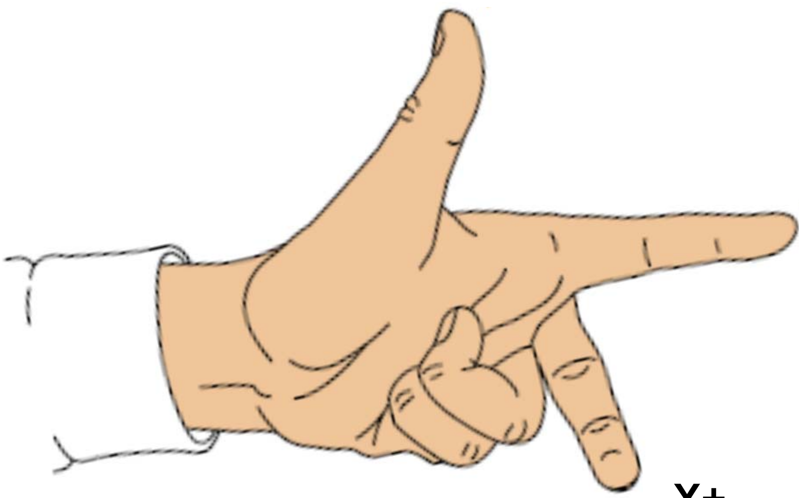

[1]

**Right-Hand Rule**

[1] - Gary Bertoline, Eric Wiebe et al, "Technical Graphics Communication", McGraw-Hill Science/Engineering/Math, 4 edition, January 31, 2008, p 310.

# Coordinate Systems

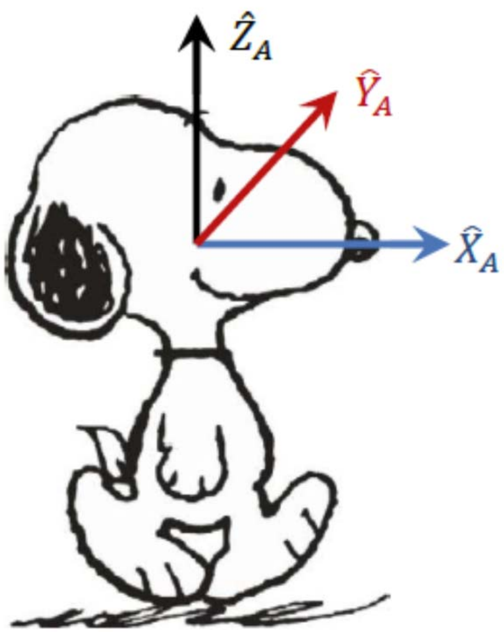
## Right Hand Rule




**Left Hand Rule (Not Documented)**

# Coordinate Systems

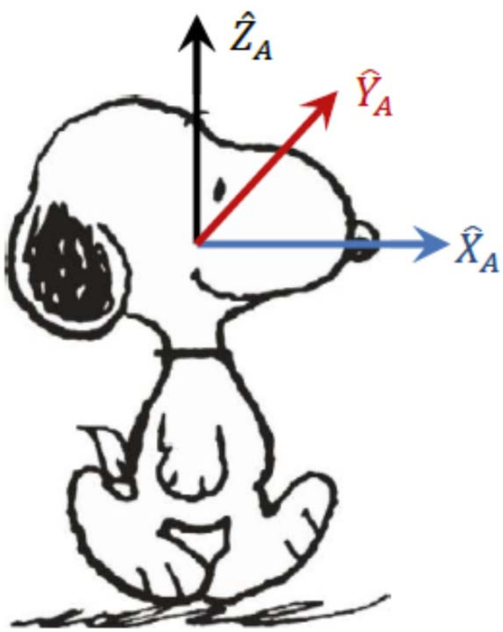
## ROTATION ABOUT AXIS



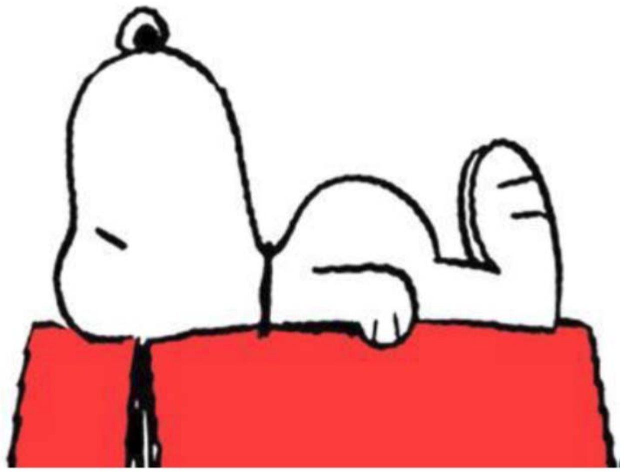

**If Snoopy is rotated 90° around Z+ axis which one of the two diagrams show its possible orientations**

# Coordinate Systems

## ROTATION ABOUT AXIS




**If Snoopy is rotated 90° around Y+ axis which one of the two diagrams show its possible orientations**

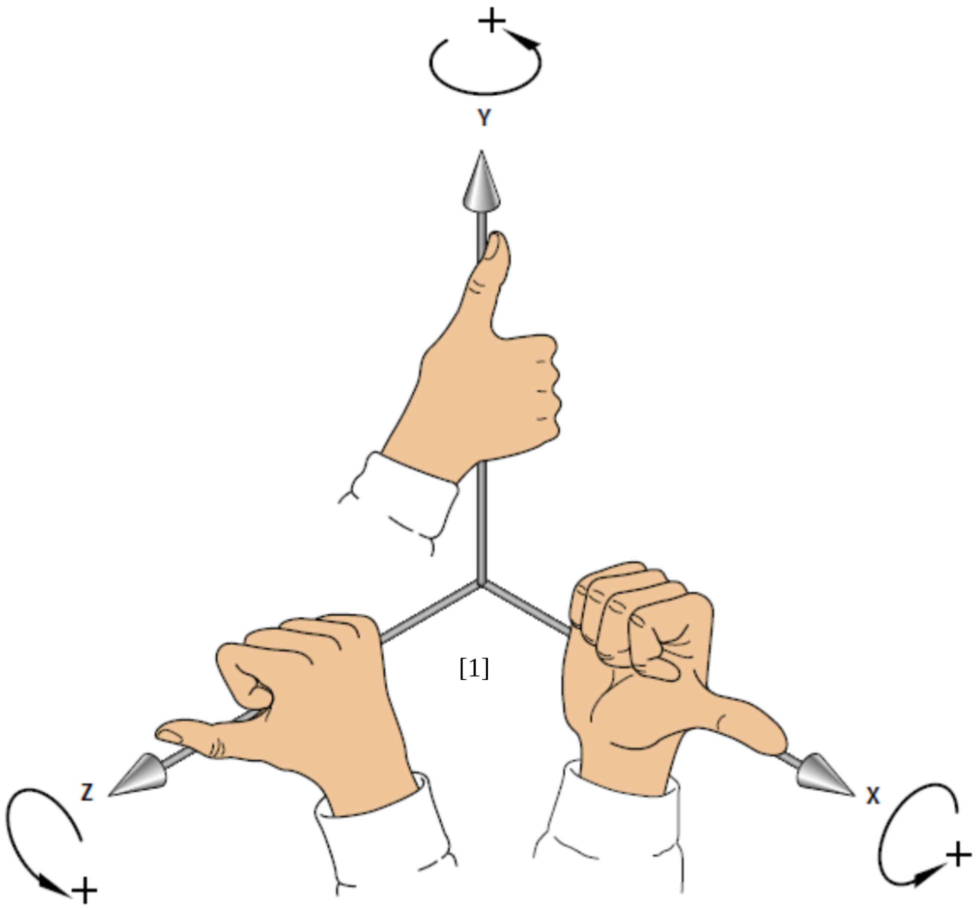


**How do you know for sure?**
**What is the rule that you are applying?**

**Right Hand Rule**

# Coordinate Systems

## Rotation About Axis – Right Hand Rule




**Positive directions of rotation on each axis.**

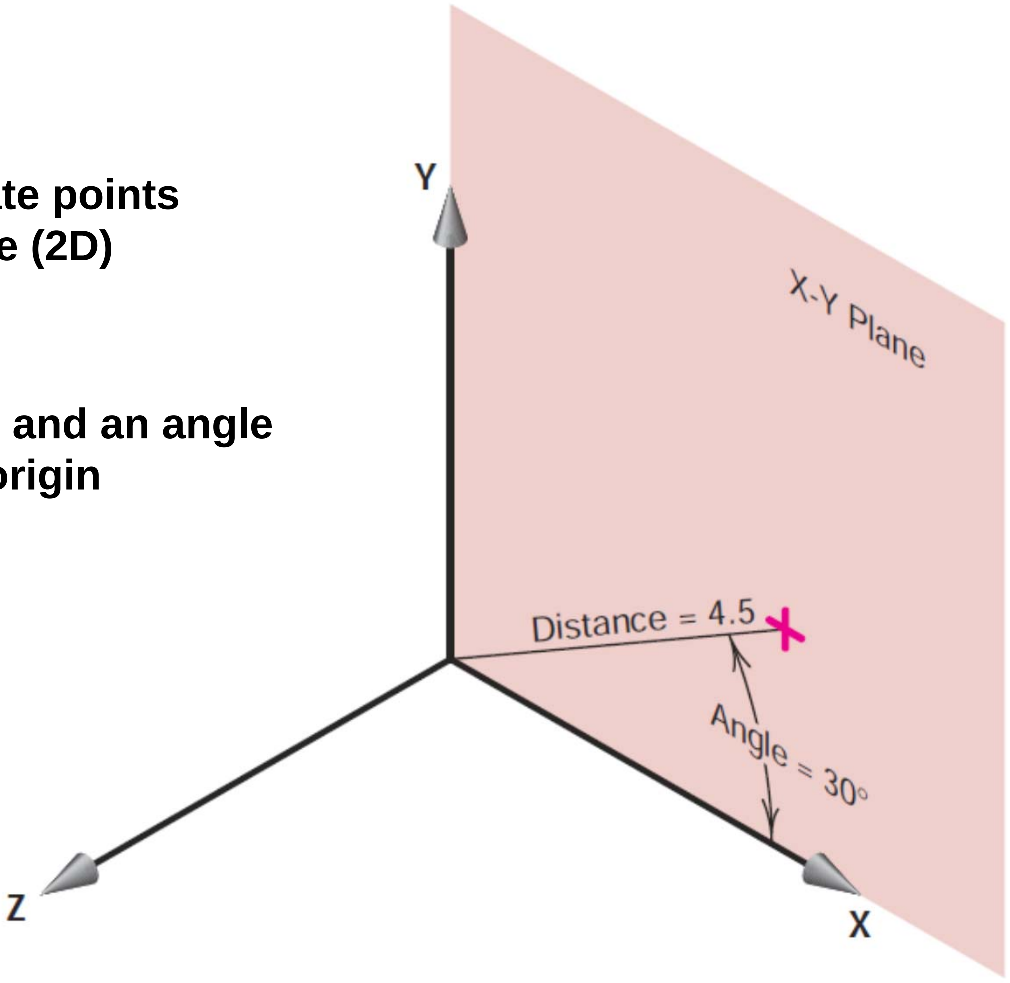
[1] - Gary Bertoline, Eric Wiebe et al, "Technical Graphics Communication", McGraw-Hill Science/Engineering/Math, 4 edition, January 31, 2008, p 310.

# Coordinate Systems

## POLAR COORDINATES

**Is used to locate points in any plane (2D)**

**Specify a distance and an angle from the origin**



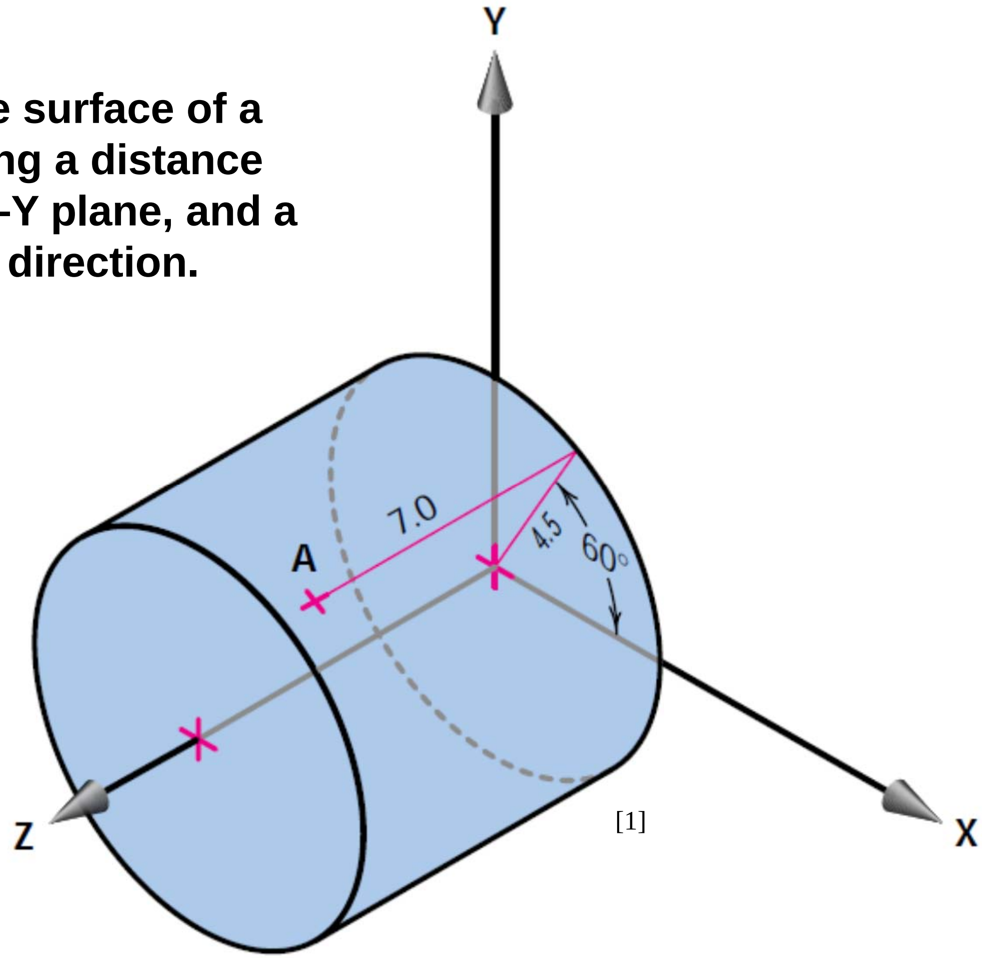

[1]

[1] - Gary Bertoline, Eric Wiebe et al, "Technical Graphics Communication", McGraw-Hill Science/Engineering/Math, 4 edition, January 31, 2008, p 311.

# Coordinate Systems

## CYLINDRICAL COORDINATES

**locate a point on the surface of a cylinder by specifying a distance and an angle in the X–Y plane, and a distance in the Z direction.**



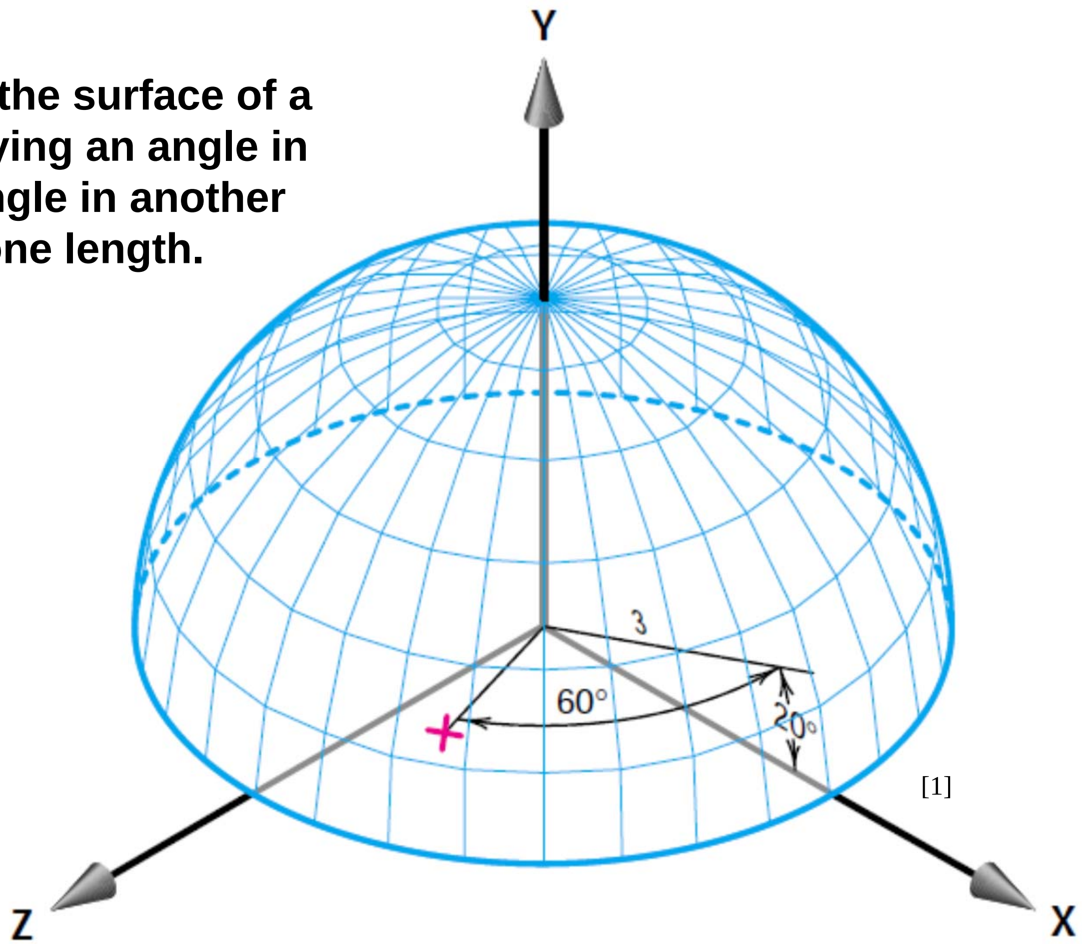

[1]

[1] - Gary Bertoline, Eric Wiebe et al, "Technical Graphics Communication", McGraw-Hill Science/Engineering/Math, 4 edition, January 31, 2008, p 311.

# Coordinate Systems

## SPHERICAL COORDINATES

**locate a point on the surface of a sphere by specifying an angle in one plane, an angle in another plane, and one length.**




[1]

[1] - Gary Bertoline, Eric Wiebe et al, "Technical Graphics Communication", McGraw-Hill Science/Engineering/Math, 4 edition, January 31, 2008, p 312.

# Take-Home Message

## CONCEPT

**Right Hand Rule**